



MMA Companion

Computer Science NEA

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Justification of features

Here I've listed out each objective/subproblem and said if they are included or not in the prototype.

Feature	Included?	Justification
<u>Objective 1: The system will generate ELO scores for every fighter inputted by a system admin using the Glicko 2 model and output a ranked leaderboard (the ELO stuff will be done when the admin updates the table)</u>	Yes	Included because it's a core function of the program.
<u>The leaderboard will display the fighter's name, rank number, record, win/loss streak, weight class and methods of winning (number of subs/decisions/kos). (Whenever I refer to displaying the table, this is the general process.)</u>	Partial	Name, record, weight class are shown. Methods and numbers are shown on the profile instead and the win/loss streak hasn't been included yet because it's a very user-deductible quality anyway which isn't as necessary as methods (By user deductible, I mean you could literally just look at the fight history whilst with the methods you would have to total everything up)
<u>The table will be sortable by all displayed factors, as well as weight class, nationality, and the belts they've held</u>	No	This objective is a pretty big UI problem because I would have to play around a bit with QT designer in order to get this to work well, and this prototype is just focused on the user and admin use-case so I had to prioritise other features
<u>On the user and admin side, you will be able to search for a fighter's name using a Levenshtein (edit based) distance system.</u>	Yes	Fighter names can be often misspelled, and users shouldn't need to type the names exactly. This is a crucial display-based objective I have to demonstrate.
<u>5. On the user and admin side, you will be able to click on a fighter's row and then see their profile come up</u>	Yes	Included because it is crucial to demonstrate the detailed stats that the system can store about each fighter and the graphics it can display.

<u>6. The fighter profile will feature a photo of the fighter, their past fight history, win streak, upcoming fights, statistics (wins/losses by KO, submission, decision) and belts</u>	Doesn't include upcoming fights, but the rest are included	The prototype doesn't include upcoming fights because they aren't necessary to have in a system which is meant to show the stats based off events that have already happened.
<u>The fighter profile will also show a pie chart that shows wins to losses and then decomposes to subs/kos/decisions to help users visualise the info.</u>	Yes	Included to show the visualisation of fighter performance and evaluate usability.
<u>The fighter profile will also show a table that lists the wins that the fighter has against the people who are currently ranked in the top 10 of their division.</u>	No	In practise with QT, the fight history on the UI isn't too large, so you would be able to spot a notable .
<u>The system must take an input of two fighters and give the odds of either of them winning by putting the ELO score they have through the Bradley-Terry model.</u>	Yes	Proves the system can take ELO data and create things like odds.
<u>Upon first use, the system must prompt the user to make a profile so they can answer a short survey about what they find entertaining about an MMA fight.</u>	Yes	Included to demonstrate how preference entry works so later features can work with data.
<u>Admins will be able to manually input fighters and fights</u>	Yes	Data entry is crucial for testing everything and proves the admin can manage the database through the UI
<u>Admins will also have the option to use a UFC scraper.</u>	No	Scraping feature adds more unnecessary dependencies to add data which I can just generate already. Not including it keeps testing more reliable

<u>The system will validate inputs by rejecting duplicate data and incorrectly formatted records</u>	No	Not included because not all the forms/fields are complete yet and the prototype is more focused on evaluating usability
<u>Users will be able to vote on fighters and give them a percentage approval rating</u>	Yes	Demonstrates a core user interaction in the system
<u>When the user calculates fighter odds, the system will give the fight a score depending on how entertaining the user finds it.</u>	No	This is a function designed for a lot of data, so I am going to implement this through testing to also refine the model I'm using more thoroughly.
<u>The UI needs to be two-toned and quite saturated to make everything easily readable (be accessible for colour-blind people if possible)</u>	Yes	Demonstrates a robust interface to help with evaluating usability
<u>When the admin makes an update on a table, the program will check for statistical anomalies</u>	No	I haven't included this because the prototype is focused on evaluating the usability of the program and the methods of access for the system by the user
<u>The users/admins will be able to navigate between all pages</u>	Yes (wording is a bit bad)	In the final program, what should really happen is the user can navigate just their allowed pages, and the admin can navigate all the pages, but currently the user is able to see the admin pages to make testing easier.
<u>The admin will be able to edit belts and events from a list using a small pop-up</u>	No	Editing Belts and events don't prove the concept of the main user journey and it doesn't add anything else of note to the prototype to add them because I can just directly put the data onto the database
<u>If there is a general exception or error, it will be displayed using a pop-up.</u>	No	I am using a terminal right now for testing purposes as I can tie the errors to print messages.
<u>The belt will be automatically allocated to the winner of the last declared title fight</u>	No	This sub-system depends on having a fully validated lineage which I haven't implemented yet.
<u>Admins will be able to jump to the leaderboard and back from the modification of the fighter list page</u>	Yes	Demonstrates the usability of the admin part by making it easy for the admin to show the result of their data population

Prototype

Prototype is showcased on the file named Prototype.mp4

Evaluation of current system

Overall, the prototype I have made meets the main objectives design and usability wise. The core case of the user (account creation to survey to leaderboard to fighter profile to odds calculator) works well, and it proves the fact that the system can store fighter data well and present it to the user without the user having to learn how the system works first.

A good strong point of the system I have right now is that the admin side is separated from the user side (even though I let the user see the admin pages at the moment for testing) but there is still a strong bridging point in the “view as user button” which makes it so the admin can populate the system with data quickly and then see if the data works on the user side, which makes testing a lot more reliable.

The current account handling system I have right now is also quite good for a prototype. The login gives you a clear error message when you type in the wrong password or username, or if you try to sign up using a username that already exists. When the user signs up it sends them straight into the survey, which is important because the whole “personalisation timeline” (survey to entertainment scores) relies on the data existing at early on.

The survey is simple but robust. Using 5 clickable stars keeps it fast and easy to use and gives the system a consistent set of values to rely on for entertainment scoring. This kind of input proves that the preference entry subsystem is usable.

The data population subsystems given to the admin are some of the strongest areas of the prototype because it makes everything else in the prototype possible. You have a completely editable fighter list. You can then go into each fighter’s profile and edit all their information and their record in detail. This part also includes tooltips for the opponent ids. This makes it so data integrity is preserved while making the program more people-friendly and readable at the same time.

The search system is very effective for the user. Instead of just searching for an exact match in the fighter list, it uses a Levenshtein edit-based distance system. This matters because when you misspell a fighter’s name, which happens a lot, you can still find the closest possible result to your search term.

The fighter profile page is one of the most valuable parts of the prototype because it demonstrates how the system can store the detailed statistics given by the admin and turn them into something that the end-user can easily understand. The page shows the record broken down into totals, and then graphically represented by a pie chart. The pie chart helps users understand the data faster by reducing the time it takes for them to process the data in their mind.

The odds calculator is also a good user feature. The entry is compact and guided using features like weight-class filtering. It shows how not only can the system display the data it has been populated with, but it also shows that the system can make mathematical inferences based off the data it has.

From a UI perspective, the two-tone colour palette makes the system more readable and consistent. This is to increase the usability of the system by making the learning curve of the UI as small as possible.

Finally, the prototype also includes features in the processing which makes the system more secure and reliable. The accounts system handles passwords that are hashed and salted, and the SQL

interactions done in the system are parametrised which prevents SQL injection techniques being used on the input fields. I am not focusing on this for the prototype, but it shows that the project should fundamentally be built on data handling.

Shortcomings and improvements

Even though the prototype proves that the main principles and ideas of the project work, there are a lot of limitations that would matter if the project were to be deployed in the state that is in right now.

Firstly, the odds calculation system doesn't take in the full potential of all the information it has as in the system you also have things like volatility and rating deviation to represent the certainty that the system has in its output, and it also doesn't consider other variables such as inactivity. This means that the percentage isn't as reliable as it can be given the info it has. To improve on this not only am I going to properly calculate the ELO system in house, but I am also going to also make a better algorithm that considers these other factors.

Secondly, I need to add a few more stats to the fighter profile to show all that the stats can tell you. This means adding things like win streaks, a proper belt system and a notable fights system. In the next iteration I am going to implement all of these.

Validation is a big part of the project which I need to improve on. Currently, a lot of the inputs can be left invalid which can break pages and mess up the code. This is obviously a big problem so an improvement to make would be to not only get the validation sweep I want up and running but also have more front-end validation and constraints on the SQL side.

Developing on the need for the validation sweep, there is currently no enforcement on foreign keys. This messes up unresolved dependencies such as records from fighter fights which don't exist. To improve on this, I need to not only add the sweep, but I can also add on constraints on the SQL side.

Finally, I need to make sure I reinforce the access control. This includes locking admin-only operations behind the admin check and making sure actions like deleting things have confirmations before you do them.